[2] I have heard it remarked in Shropshire that the water, when the Severn is flooded from long-continued rain, is much more turbid than when it proceeds from the snow melting in the Welsh mountains. D'Orbigny (tom. i. p. 184), in explaining the cause of the various colours of the rivers in South America, remarks that those with blue or clear water have there source in the Cordillera, where the snow melts.

[3] Dr. Gillies in Journ. of Nat. and Geograph. Science, Aug., 1830. This author gives the heights of the Passes.

[4] This structure in frozen snow was long since observed by Scoresby in the icebergs near Spitzbergen, and, lately, with more care, by Colonel Jackson (Journ. of Geograph. Soc., vol. v. p. 12) on the Neva. Mr. Lyell (Principles, vol. iv. p. 360) has compared the fissures by which the columnar structure seems to be determined, to the joints that traverse nearly all rocks, but which are best seen in the non-stratified masses. I may observe, that in the case of the frozen snow, the columnar structure must be owing to a "metamorphic" action, and not to a process during deposition.

[5] This is merely an illustration of the admirable laws, first laid down by Mr. Lyell, on the geographical distribution of animals, as influenced by geological changes. The whole reasoning, of course, is founded on the assumption of the immutability of species; otherwise the difference in the species in the two regions might be considered as superinduced during a length of time.

CHAPTER XVI

NORTHERN CHILE AND PERU

Coast-road to Coquimbo—Great Loads carried by the Miners—Coquimbo —Earthquake—Step-formed Terrace—Absence of recent Deposits—Contemporaneousness of the Tertiary Formations—Excursion up the Valley —Road to Guasco—Deserts—Valley of Copiapo—Rain and Earthquakes—Hydrophobia—The Despoblado—Indian Ruins—Probable Change of